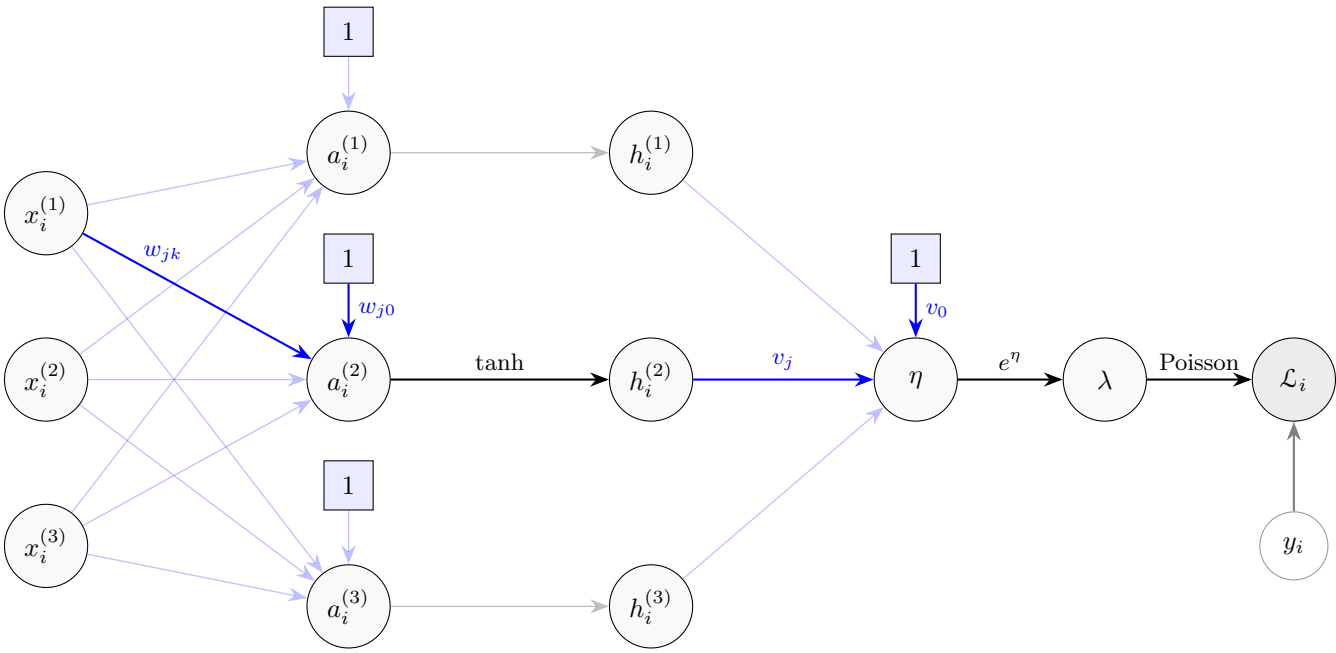
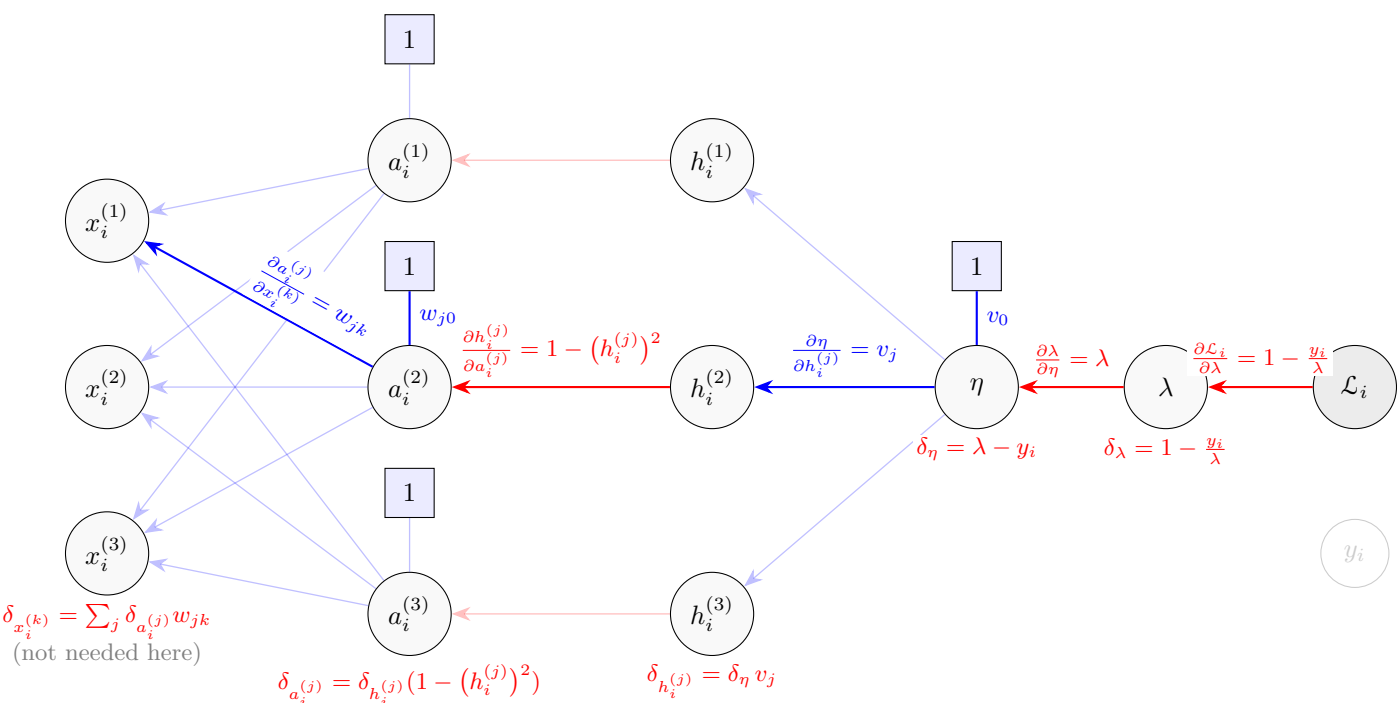
forward pass: values



backward pass: local derivatives



$$\frac{\partial \mathcal{L}_i}{\partial w_{jk}} = \delta_{a_i^{(j)}} \, x_i^{(k)}$$

$$\frac{\partial \mathcal{L}_i}{\partial w_{j0}} = \delta_{a_i^{(j)}} \cdot 1 = \delta_{a_i^{(j)}}$$

$$\frac{\partial \mathcal{L}_i}{\partial v_j} = \delta_\eta \, h_i^{(j)}$$

$$\frac{\partial \mathcal{L}_i}{\partial v_0} = \delta_\eta \cdot 1 = \delta_\eta$$